

Multiple Choice (10 marks)

1. Assessing growth in routine practice should be:
 - (a) Height only
 - (b) Weight only
 - (c) Weight and height
 - (d) Head circumference
2. The function of the gastrointestinal tract can be classified in several ways. Which of the following functions is most important for survival immediately after a meal?
 - (a) Absorption and retention of water and electrolytes
 - (b) Elimination of toxins
 - (c) Maintenance of barrier function
 - (d) Extraction of maximum value from nutrients
3. Which of the following is the yield of ATP from complete oxidation of glucose to carbon dioxide and water?
 - (a) 10
 - (b) 25
 - (c) 30
 - (d) 32
4. Which of the following statements is correct?
 - (a) Cholesterol mainly occurs in the cell walls of bacterial cells
 - (b) Cholesterol mainly occurs in the cell walls of mammalian cells
 - (c) Cholesterol mainly occurs in the cell walls of yeast cells
 - (d) Cholesterol mainly occurs in plant cell walls.
5. Which of the following is a function of folate?
 - (a) Synthesis of TMP (thymidine monophosphate)
 - (b) Decarboxylation of amino acids to form amine neurotransmitters
 - (c) Synthesis of methionine from homocysteine
 - (d) Carboxylation of pyruvate to oxaloacetate

True / False (10 marks)

Answer True or False

6. True or False: Vitamin C is essential for providing the cofactor necessary for hydroxylation reactions.
7. True or False: A food additive is safe only if it shows no toxic effects at any dose during its entire lifetime.
8. True or False: Dietary fat provides energy, forms part of cellular membranes, and plays a role in cell signaling.
9. True or False: The Atherogenic Lipoprotein Phenotype (ALP) is characterized by raised levels of LDL cholesterol and triglycerides, and lowered levels of HDL cholesterol.
10. True or False: In type 2 diabetes, hyperglycemia is primarily caused by increased glucose uptake by cells.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the methods used to evaluate protein quality in human nutrition and explain why animal growth assays may not accurately reflect dietary protein quality for humans.
12. Discuss the various beneficial effects of exercise for individuals with diabetes, and critically analyze the evidence for these effects.

End of Paper